Arctic Sea Ice and Open Water Classification From Spaceborne Fully Polarimetric Synthetic Aperture Radar

Yiru Lu, Biao Zhang, *Senior Member, IEEE*, and William Perrie

***Abstract*— We present an approach for automatic classification of Arctic sea ice and open water from spaceborne C-band RADARSAT-2 fully polarimetric (HH, HV, VH, VV) synthetic aperture radar (SAR) imagery based on the random forest (RF) model. The HH- and HV-polarized radar backscatter, incidence angle, and optimal polarimetric features are inputs of the RF model. First, we use a physics-based unsupervised scheme to generate well-annotated sea ice and open water samples. This scheme can alleviate labeling errors arising from visual interpretation or temporal–spatial mismatch between SAR images and operational ice charts. Subsequently, we estimate a set of characteristic polarimetric parameters and select the most representative features using the recursive feature elimination method. Finally, the RF model is trained and validated using more than one million labeled samples. Statistical validation results show that the overall sea ice and water classification accuracy is 99.94% and the Kappa coefficient is 0.999. We find that ice–water discrimination accuracy can be improved by about 4%–10% when optimal polarimetric features are involved in the RF model input. Moreover, the polarization difference (PD, VV–HH) is found to be the foremost polarimetric parameter for distinguishing sea ice from open water. The proposed approach has the capability of yielding satisfactory ice–water classification over the complicated marginal ice zone. High-resolution ice–water classification maps clearly show that sea ice leads and their surroundings are also well separated.**

***Index Terms*— Classification, quad-polarization, random forest (RF), sea ice, synthetic aperture radar (SAR).**

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I. Introduction

ARCTIC sea ice plays an extremely crucial role in the Earth's climate system and has a significant influence on global sea level, atmospheric and oceanic circulations, global surface energy, and moisture budgets [1]. The large-scale reduction in Arctic sea ice in recent decades is considered as one of the most representative indicators for global climate change. Satellite observations have revealed the remarkable decline trend in Arctic sea ice extent, thickness, and volume [2], [3], [4], [5], [6], [7], predominately caused by strong natural variability in the coupled ice–ocean–atmosphere system and anthropogenic carbon dioxide emissions [8], [9]. For a better understanding of the ongoing Arctic sea ice changes and their relation to the local and global climate, a more detailed and accurate knowledge of sea ice extent is necessary. Furthermore, decreases in sea ice cover in the Arctic will lead to more frequent polar shipping and transport, as well as offshore oil and gas exploration. Consequently, high-resolution and timely sea ice extent information is imperative to support safe navigation and resource exploitation activities in the Arctic.

Satellite microwave sensors, such as radiometer or scatterometer, can provide long time series and large coverage of Arctic sea ice concentration and extent [10], [11]. Yet, they are not able to resolve small-scale features (e.g., leads) or to provide accurate sea ice cover in the complicated marginal ice zone (MIZ), due to their coarse spatial resolutions. Moreover, large uncertainties may exist in radiometer-derived sea ice concentration estimates during melting seasons. Synthetic aperture radar (SAR) is a unique sensor to monitor polar regions because it is not affected by cloud or illumination and can provide observations under all-weather and day and night conditions, even during the polar night. Spaceborne C-band SARs, such as RADARSAT-2 (RS-2), Sentinel-1A/B, and Radar Constellation Mission (RCM), can provide high-resolution and wide swath Arctic sea ice observations and thus have the potential to yield more accurate and detailed sea ice extent information than is possible from radiometer or scatterometer sensors.

Single- or dual-pol SAR data have been widely used to discriminate sea ice and open water based on statistical approaches (e.g., local autocorrelation and adaptive probability thresholding) [12], [13], conventional machine learning algorithms (e.g., logistic regression, decision tree,

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artificial neural network, and support vector machine) [14], [15], [16], [17], [18], [19], or advanced deep learning models (e.g., U-Net and a fully convolutional network) [20], [21]. Sea ice extent and ice concentration can be directly determined once sea ice and open water are discriminated using the aforementioned methodologies. A recent study has shown that the assimilation of SAR-derived ice–water classification results in a regional ice–ocean prediction system significantly improves the ice concentration analysis [22].

Regarding the data–driven methods for SAR ice–water classification, the most important procedure is to create a high-quality training dataset. To our knowledge, an accurate ice–water segmentation reference ("ground truth") and a large number of SAR images are two key factors to generate well-annotated and generalized training samples. Ordinarily, the ice–water segmentation reference is based on two component activities: 1) visual interpretation of SAR images [13], [17], [20] and 2) manually derived sea ice charts from operational ice service divisions [15], [16]. The disadvantage of visual interpretation is that it is time-consuming and tends to yield inaccurate label results, especially for small objects (e.g., leads and ice floes). Moreover, visual inspection also leads to mislabeling when sea ice and open water have very similar backscattering characterizations, which can occur under certain incidence angle and wind conditions.

Daily ice charts are produced from an integrated analysis by sea ice analysis experts, using available optical and microwave satellite data as well as in situ observations from coastal stations and ships. However, ice charts include operator biases caused by subjective analysis and representative errors. Based on colocations of ice charts and SAR images, one can determine whether a particular pixel in a SAR scene represents sea ice or open water. However, the time difference between the SAR image acquisition and the ice chart may induce analysis error, especially for highly dynamic sea ice regions (e.g., Fram Strait), because the sea ice can rapidly move and modulate into a myriad of different ice types, thereby undergoing significant change over a time scale of hours. In addition to visual interpretation of SAR images and information from sea ice charts, a regional-based image segmentation approach, called watershed transformation, has been used to obtain training samples for extracting sea ice cover information from SAR imagery [18]. However, this method inevitably generates a small quantity of examples of sea ice mixed with open water.

For single- or dual-pol SAR, radar backscattering intensity, incidence angle, and also texture parameters are routinely used as inputs to various algorithms to discriminate sea ice and open water [12], [13], [14], [15], [16], [17], [18], [19], [20], [21]. Although these methods are useful and successful in certain conditions, they still face major challenges in order to obtain more accurate ice–water classification results, especially at the ice–water boundary and for small-scale features within pack ice. An early study has previously shown that ice–water classification from single-pol SAR images is still ambiguous in some cases [23]. These obstacles might be solved by involving multipol SAR observations.

Fully polarimetric SAR systems measure the scattering matrix of the object, providing both backscattering intensity and phase information from the four orthogonal polarization combinations (HH, HV, VH, and VV). The complex elements of the scattering matrix can be used to estimate a set of polarimetric features (polarization ratio, co-polarization phase difference, entropy, and so on). These characteristic polarimetric parameters are capable of describing and quantifying differing scattering mechanisms, which have been used to discriminate different sea ice types [24], [25], [26], [27], [28]. A previous study has demonstrated that the accuracy and efficiency of sea ice classification can be improved by utilizing polarimetric parameters [25]. In addition to sea ice classification, only a few studies have reported sea ice and open water discrimination from fully polarimetric SAR imagery [29], [30]. Sea ice can be separated from open water based on discrepant scattering mechanisms. For example, multiyear ice and open water correspond to volume scattering and surface scattering, respectively.

Leads in the sea ice develop as quasi-rectilinear cracks within the pack ice due to sea ice dynamics and warm water upwelling and vary from a few meters to hundreds of meters in width [31], [32]. Small changes in the fraction of leads in a given area have the potential to induce large temperature fluctuations in the atmospheric boundary layer [33], thereby affecting the energy exchange between ocean and atmosphere. Thus, leads are of interest for climate studies, weather forecasting, and ship navigation in polar regions. Leads can be detected from single-pol (e.g., HH or VV) SAR images but are possibly misclassified as sea ice under high wind conditions. This issue is able to be alleviated by utilizing dual-pol (e.g., HH and HV) SAR data [34]. Fully polarimetric SAR provides simultaneous co-pol (HH, VV) and cross-pol (HV, VH) observations. The use of four-channel SAR images instead of one enhances the capability to characterize the scattering process. Moreover, the noise-equivalent sigma zero (NESZ) of fully polarimetric SAR is much lower than that of single- or dual-pol SAR. These unique characteristics give fully polarimetric SAR the potential to correctly identify leads.

Fully polarimetric SAR has high spatial resolution, but its swath is relatively small (25–50 km) due to the doubled pulse repetition frequency and the alternate transmission of orthogonal linear polarization (e.g., H and V) pulses. Therefore, this inherent technical limitation constrains its operational utility for sea ice monitoring. To overcome the dilemma between wide unambiguous swath coverage and high azimuth resolution for spaceborne SAR systems, innovative techniques, such as high-resolution wide-swath (HRWS) imaging and the staggered SAR concept, have been developed and proposed to acquire fully polarimetric observations with large coverage [35], [36], [37], [38], [39], based on digital beamforming and multiple aperture signal recording. Future SAR satellites will have the capability to achieve high-resolution (5 m) fully polarimetric imaging over wide swath (400 km), providing an unprecedented opportunity for remote sensing of sea ice with an operational perspective. Moreover, the

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compact polarimetric (CP) imaging mode provides a compromise between the abundant polarimetric information of quad-pol SAR systems and large coverage of standard single- and dual-pol SARs [40], [41]. CP SAR transmits a right-circular polarization and receives two coherent orthogonal linear polarizations (RH and RV). RCM has CP imaging mode and can offer multilook complex images with a medium resolution of 50 m and large swaths of 350 km. CP SAR data have demonstrated their usefulness for operational sea ice classification [42], [43], [44], [45].

This study aims to develop an automatic classification algorithm for sea ice and open water using spaceborne fully polarimetric SAR. We conduct a comprehensive evaluation to demonstrate the reliability of the proposed method. In particular, we seek to answer specific research questions as follows: 1) how can we generate unsupervised training samples? 2) can we improve ice–water classification accuracy by incorporating a variety of polarimetric features in the classification model? and 3) what is the most representative polarimetric parameter for distinguishing sea ice from open water? The remainder of this article is organized as follows. Section II describes the datasets. In Section III, the SAR image preprocessing, generation of training samples, feature extraction and selection, and ice–water classification method are introduced. In Section IV, we present the results and validation. Finally, a summary and the conclusions are given in Section V.

II. Datasets

A. RS-2 Quad-Polarization SAR Data

In this study, we collected a total of 869 C-band RS-2 fine quad-pol (HH, HV, VH, VV) SAR images, from October 2008 to September 2019, to develop and validate the proposed automatic ice–water classification algorithm. Among these, 608 SAR scenes of both sea ice and open water were acquired over the Arctic, and the remainder was open water observations over the Gulf of Alaska, East and West Coasts of USA. These images were selected to cover various sea ice regimes (e.g., young ice, first-year ice, and multiyear ice) and open water, under different wind and sea state conditions (e.g., rough and calm water, and ice leads). Fig. 1 shows the geographic locations of RS-2 quad-pol SAR images acquired over the Arctic. For quad-pol imaging mode, the swath width is 25 km. The incidence angles range from 18.5° to 49.5°, and the spatial resolutions in the range and azimuth directions are 5.4 × 8 m, respectively. The NESZ is about −36.5 ± 3 dB [46].

B. Global Self-Consistent Hierarchical High-Resolution Geography Database

The Global Self-consistent, Hierarchical, High-resolution Geography (GSHHG) Database (https://www.soest.hawaii.edu/pwessel/gshhg/) is used to mask the land present in the RS-2 quad-pol SAR images. The binary shoreline data were recorded in a standard NetCDF file format, and the resolution is 1 × 1 arc-min for the latest GSHHG (version 2.3.7).

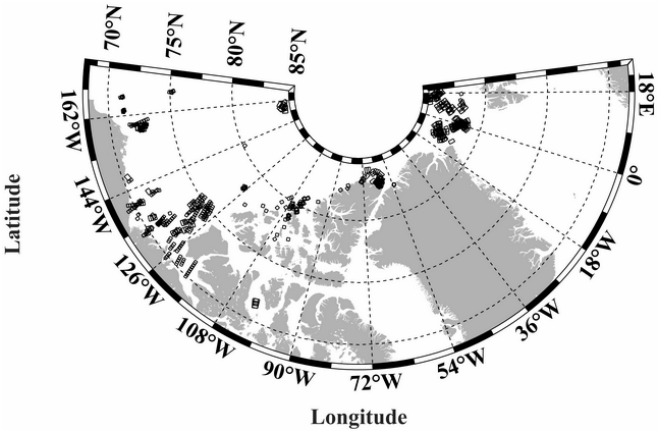


Fig. 1. Geographic locations of RS-2 quad-polarization SAR imagery acquired over the Arctic ocean.

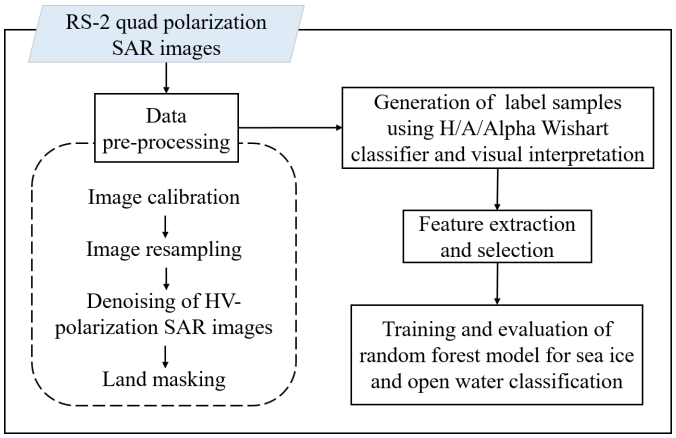


Fig. 2. Flowchart of the proposed method for classifying sea ice and open water from RS-2 quad-polarization SAR data.

III. Methodology

In this section, we present a detailed description of the proposed method for classifying sea ice and open water from RS-2 fine quad-pol SAR imagery. The procedure for ice–water classification consists of SAR image preprocessing, the generation of training samples, feature extraction and selection, discrimination of sea ice from open water using the random forest (RF) model, and evaluation of the classification results. The flowchart for the algorithm is shown in Fig. 2.

A. SAR Image Preprocessing

Prior to ice–water classification, we need to conduct SAR image preprocessing, including absolute calibration to obtain the normalized radar cross section (NRCS) for each image, speckle noise removal, thermal noise correction, and land masking. For each RS-2 fine quad-pol SAR image, the range of incidence angle is about 1.5° from the near range to far range. HH- or VV-pol radar backscatter does not significantly vary over such a small incidence angle range. Moreover, previous studies have shown that the cross-pol (HV or VH) radar returns from open water are not sensitive to incidence angles, especially under high wind conditions [47], [48], [49]. Thus, in this study, the correction of incidence

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angle dependence on quad-pol radar backscatter is not needed. However, it should be noted that incidence angle correction is necessary for classification of sea ice types and discrimination of sea ice from open water using single- or dual-pol SAR wide swath images, acquired over large incidence angle ranges [50], [51], [52], [53], because both HH- and HV-pol radar backscatters from sea ice show strong angular dependence [51].

We first implement the radiometric calibration on RS-2 quad-pol SAR single complex look (SLC) data using the following formula:

$$\text{Calibrated value} = \frac{|\text{digital value}|^2}{A^2} \tag{1}$$

where the calibrated value (real number) is HH-, HV- or VH-, and VV-pol NRCS (linear unit), the digital value is the polarimetric scattering coefficient (complex number) in the scattering matrix, and A is the gain value corresponding to the range sample. The elements in scattering matrix are used to estimate various polarimetric features.

Speckle noise in SAR images degrades the data quality and may affect the ice–water classification results. Therefore, we apply a boxcar average operation on the original RS-2 quad-pol SAR images to reduce noise and smooth the images to 100-m pixel spacing. To reduce the thermal noise effect, NESZ values are subtracted from the HV-pol NRCS values. Finally, the GSHHG database is used to mask the land areas in quad-pol SAR images.

B. Generation of Training Samples

Due to the lack of available in situ sea ice observations in the polar regions, visual interpretation of SAR images and manually derived ice charts by sea ice experts are common ways used to label sea ice and open water samples. The annotated samples are further utilized to train and validate various machine learning models for ice–water classification with single- and dual-pol SAR data. However, the labeled samples generated from the aforementioned two approaches inevitably contain label errors due to subjective analysis by the operator, as well as the time interval and spatial resolution discrepancies between SAR images and the ice charts.

In this study, for the purpose of generating training samples, we first apply an unsupervised method to automatically segment quad-pol SAR images acquired in the Arctic. This segmentation method is referred to as the H/A/Alpha–Wishart classifier, which is a synthesis of the unsupervised target decomposition classifier [54], [55], [56] and the supervised Wishart classifier [57]. This method has been used for terrain classification from multifrequency and fully polarimetric SAR images [58]. The H/A/Alpha–Wishart classifier is linked to physical scattering properties. No human intervention is involved in this segmentation process.

Fig. 3(a)–(c), respectively, shows RS-2 VV-, HH-, and HV-pol SAR images acquired in a fine quad-pol mode over the Beaufort Sea on July 8, 2017 at 17:31 UTC. The incidence angles range from 30.4° to 32.0°. As an example, we use the H/A/Alpha–Wishart classifier to segment this SAR image. The

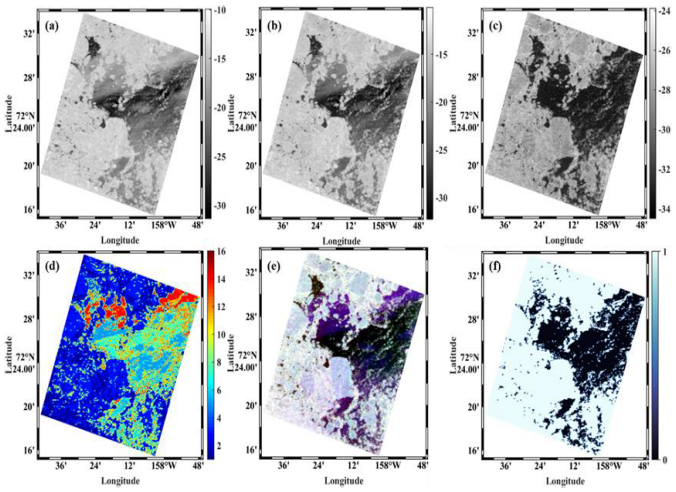


Fig. 3. (a) VV-, (b) HH-, and (c) HV-polarized RS-2 SAR images acquired over the Beaufort Sea on July 8, 2017 at 17:31 UTC. The color bar represents NRCS in VV-, HH-, and HV-polarization, in units of decibels. (d) SAR image segmentations from the H/A/Alpha–Wishart classifier. The color bar denotes 16 classes. (e) Composite Pauli RGB image, with red (|HH + VV|), green (|HV|), and blue (|HH – VV|). (f) Ice–water classifications from the proposed algorithm. The light cyan and dark in the color bar stand for sea ice and open water, respectively. RS-2 Data and Product MacDonald, Dettwiler, and Associates Ltd., all rights reserved.

segmentation consists of three procedures. First, we estimate entropy (H), alpha angle (α), and anisotropy (A) from an eigenvalue decomposition of the coherency matrix based on the polarimetric target decomposition theorem [59]. Estimation of these three parameters allows a classification of the SAR image because they are relevant to the underlying scattering process (H and A) and physical scattering mechanism (α). Subsequently, we initially segment the SAR image using the H/A/Alpha classifier. Regarding this scheme, the H/A/Alpha 3-D space is divided into 16 zones and each zone is associated with the general properties of the scattering mechanism. The physical scattering characteristics relating to each zone provide information for clustering the same type of target. Finally, we estimate the initial cluster center of each zone based on the initial segmentation map, and the distances between each pixel and the cluster center of each zone. The details regarding the distance measurements can be found in [57]. Each pixel is assigned to a specific class based on the minimum distance criterion. The iteration operation can be implemented in the last procedure to achieve more reliable segmentations. Fig. 3(d) shows the image segmentations from the H/A/Alpha–Wishart classifier. Based on unsupervised segmentation results, we further inspect HH- and HV-pol SAR images and the associated composite Pauli RGB image [Fig. 3(e)], assigning each pixel to a specific type, sea ice, or open water. The wind-roughened open water and small ice floes are well separated, as shown in Fig. 3(f).

We apply the H/A/Alpha–Wishart classifier to segment all RS-2 quad-pol SAR images, generating a total of 1 311 609 labeled samples. Among these, 656 674 samples are randomly selected to train and test the RF model. The residual 655 935 samples are used to validate the model. The percentages of sea ice and open water samples are almost the same for

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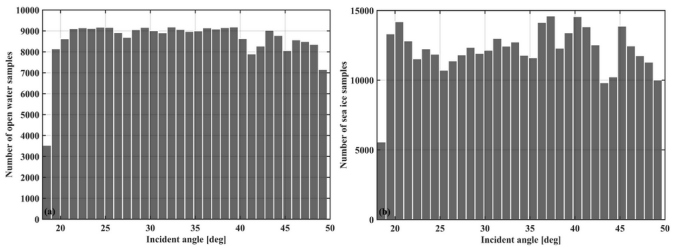


Fig. 4. Number of sea ice and open water samples versus different incidence angles.

model training and validation. We divide the incidence angles (18.5°–49.5°) of all samples into 32 bins with intervals between two adjacent bins set to 1°. For each incidence angle bin, we estimate the number of sea ice and open water samples; the results are shown in Fig. 4. The number of samples in each bin does not significantly vary with incidence angle, except for the smallest incidence bin (18.5°), whether for the sea ice or open water. As such, a balanced annotated ensemble is created.

C. Feature Extraction

In this study, we extract both polarimetric features and texture features from RS-2 quad-pol SAR data. First, 33 polarimetric features (polarization difference (PD), cross-polarization ratio, scattering mechanism entropy, and so on) are estimated from scattering matrix elements and eigende-composition parameters of the coherence matrix, according to the formulas presented in [60], [61], [62], [63], [64], [65], and [66]. The derivation of these polarimetric features is summarized in Table I. Subsequently, we estimate 18 texture features, and among these, 17 are calculated from HH- and HV-pol SAR intensity images using the gray-level co-occurrence matrix (GLCM). The GLCM technique [67] is originally developed for extracting textural properties of blocks of image data, by computing a set of gray-level spatial-dependence probability distribution matrices for various angular relationships, and distances between neighboring cell pairs in the image. More specifically, GLCM calculates the probability of a pixel with gray-level value i occurring at a certain distance and angle to adjacent pixels with gray-level value j. The definitions of texture features are given in Table II. To estimate texture features, the number of gray levels is chosen to be 32, with average orientations of 0°, 45°, 90°, and 135°, window size of 5 × 5 pixels, and co-occurrence distance of 4 pixels.

D. Feature Selection and Ice–Water Classification

We implement ice–water classification using RS-2 quad-pol SAR data based on the RF model. RF is a supervised ensemble learning method, which integrates a number of decision trees into a forest for the purpose of classification and regression [68]. Each individual tree is independently constructed and trained using various subsets of the dataset and various subsets of available features. This unique random selection of subsets of the dataset and features can avoid the common overfitting issue. The majority vote of the trees is considered as the final decision of the RF model. This criterion reduces the overall variance and bias of the RF classifier.

TABLE I

DESCRIPTION, MATHEMATICAL DEFINITION, AND RANK OF IMPORTANCE OF 33 POLARIMETRIC FEATURES (PD, γ_{cr}, H, AND SO ON) AND THREE IMAGE FEATURES (σ^0_{HH}, σ^0_{HV}, θ)

Symbol	Name	Formula	Rank of importance
PD	Polarization difference	$PD = \langle\lvert S_{VV}\rvert^2\rangle - \langle\lvert S_{HH}\rvert^2\rangle$	1
σ^0_{HV}	HV-polarization backscatter	N/A	2
γ_{cr}	Cross-polarization ratio [39]	$\gamma_{cr} = \langle\lvert S_{HV}\rvert^2\rangle/\langle\lvert S_{HH}\rvert^2\rangle$	3
H (Van Zyl)	Scattering mechanism entropy [40]	$H = -\sum_i^{n=3} p_i log_3(p_i)$, $p_i = \lambda_i/\sum_i^{n=3}\lambda_i$	4
$A12$	Anisotropy12	$A12 = \lambda_1 - \lambda_2/\lambda_1 + \lambda_2$	5
γ_{co}	Co-polarization ratio [39]	$\gamma_{co} = \langle\lvert S_{HH}\rvert^2\rangle/\langle\lvert S_{VV}\rvert^2\rangle$	6
ρ	Real part of the co-polarization cross product [41]	$\rho = \lvert\Re\langle S_{HH}S^*_{VV}\rangle\rvert$	7
θ	Radar incidence angle	N/A	8
σ^0_{HH}	HH-polarization backscatter	N/A	9
ν	Geometric intensity [41]	$\nu = (det(T_3))^{1/3}$	10
ε	Co-polarization correlation coefficients [41]	$\varepsilon = \langle S_{HH}S^*_{VV}\rangle/\sqrt{\langle\lvert S_{HH}\rvert^2\rangle\langle\lvert S_{VV}\rvert^2\rangle}$	11
H (Freeman)	Scattering mechanism entropy [42]	$H = -\sum_i^{n=3} p_i log_3(p_i)$, $p_i = \lambda_i/\sum_i^{n=3}\lambda_i$	12
$SERD$	Single bounce eigenvalues relative difference	Eq.(7.29) in [43]	13
H (Cloude & Pottier)	Entropy [35]	$H = -\sum_i^{n=3} p_i log_3(p_i)$, $p_i = \lambda_i/\sum_i^{n=3}\lambda_i$	14
PH	Pedestal height [43]	$PH = \lambda_3/\lambda_1$	15
SE	Shannon entropy [43]	$SE = \log(\pi^3 e^3\lvert T_3\rvert)$	16
$\overline{\alpha}$ (Cloude & Pottier)	Alpha [35]	$\overline{\alpha} = \sum_i^{n=3} p_i\alpha_i$, $p_i = \lambda_i/\sum_i^{n=3}\lambda_i$, $0° \le \alpha_i = \cos(\lvert u_i(1)\rvert)^{-1} \le 90°$	17
PA	Polarimetric asymmetry [43]	$PA = \lambda_1 - \lambda_2/span - 3\lambda_3$	18
δ	Scattering diversity [41]	$\delta = 3/2 \times (1 - (\lVert T_3\rVert_F/span)^2)$	19
SE_P	Polarimetric component of Shannon entropy [43]	$SE_P = \log(4\lvert T_3\rvert/\mathrm{Trace}(T_3)^2)$	20
$span$	Span of coherency matrix T_3 [39]	$span = \mathrm{Trace}(T_3)$	21
A (Lueneburg)	Lueneburg anisotropy	Eq. (7.36) in [43]	22
$\overline{\alpha}$ (Praks & Colin)	Alpha approximation	Eq. (7.49) in [43]	23
H (Praks & Colin)	Entropy approximation	Eq. (7.46) in [43]	24
$H(1-A)$	N/A	$H(1-A)$	25
PF	Polarization fraction [43]	$PF = 1 - 3\lambda_3/span$	26
$(1-H)(1-A)$	N/A	$(1-H)(1-A)$	27
τ	Surface scattering fraction [41]	$\tau = \langle\lvert S_{HH} + S_{VV}\rvert^2\rangle/span$	28
RVI	Radar vegetation index [43]	$RVI = 4\lambda_3/(\lambda_1 + \lambda_2 + \lambda_3)$	29
$\overline{\alpha}$ (An, Cui & Yang)	Alpha approximation	Eq. (7) in [44]	30
SE_i	Intensity component of Shannon entropy [43]	$SE_i = 2\log(\pi e\mathrm{Trace}\lvert T_3\rvert/2)$	31
μ	Conformity coefficient [45]	$\mu = \frac{2(\Re\langle S_{HH}S^*_{VV}\rangle - \langle\lvert S_{HV}\rvert^2\rangle)}{(\langle\lvert S_{HH}\rvert^2\rangle + \langle\lvert S_{VV}\rvert^2\rangle + 2\langle\lvert S_{HV}\rvert^2\rangle)}$	32
$(1-H)A$	N/A	$(1-H)A$	33
$DERD$	Double bounce eigenvalues relative difference	Eq. (7.29) in [43]	34
HA	N/A	HA	35
A (Cloude & Pottier)	Anisotropy [35]	$A = \lambda_2 - \lambda_3/\lambda_2 + \lambda_3$	36

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TABLE II

DESCRIPTION, MATHEMATICAL DEFINITION, AND RANK OF IMPORTANCE OF 18 TEXTURE FEATURES AND THREE IMAGE FEATURES (σ^0_{HH}, σ^0_{HV}, θ)

Name	Formula	Rank of importance	
		HH	HV
Angular second moment	$\sum_{i=1}^{k}\sum_{j=1}^{k}S(i,j)^2$	29	27
Contrast	$\sum_{i=1}^{k}\sum_{j=1}^{k}(i-j)^2S(i,j)$	7	5
Correlation	$\sum_{i=1}^{k}\sum_{j=1}^{k}S(i,j)\frac{(i-Mean)(j-Mean)S(i-j)^2}{VAR}$	15	16
Variance	$\sum_{i=1}^{k}\sum_{j=1}^{k}S(i,j)\,(i-Mean)^2$	14	11
Inverse difference moment	$\sum_{i=1}^{k}\sum_{j=1}^{k}S(i,j)$	13	35
Sum average	$\sum_{i=2}^{2k}iS_{x+y}(i)$, *where x and y are the coordinates of an entry in the co-occurrence matrix*	3	12
Sum variance	$\sum_{i=2}^{2k}(i-SENT)^2S_{x+y}(i)$	9	38
Sum entropy	$\sum_{i=2}^{2k}S_{x+y}(i)\,log_{10}[S_{x+y}(i)]$	23	30
Entropy	$\sum_{i=1}^{k}\sum_{j=1}^{k}S(i,j)log_{10}[S(i,j)]$	24	32
Difference variance	$\sum_{i=1}^{k}i^2\,S_{x-y}(i)$	20	36
Difference entropy	$-\sum_{i=1}^{k}S_{x-y}(i)\,log_{10}[S_{x-y}(i)]$	28	37
Information measures of correlation 1	$\frac{HXY-HXY_1}{\max\{HX,HY\}}$, *where HXY=ENT, HX, HY are the entropies of* S_x *and* S_y $HXY_1=-\sum_{i=1}^{k}\sum_{j=1}^{k}S(i,j)log_{10}[S_x(i)S_y(j)]$, $HXY_2=-\sum_{i=1}^{k}\sum_{j=1}^{k}S_x(i)S_y(j)log_{10}[S_x(i)S_y(j)]$	26	39
Information measures of correlation 2	$(1-\exp[-2(HXY_2-HXY)])^{\frac{1}{2}}$	31	32
Dissimilarity	$\sum_{i=1}^{k}\sum_{j=1}^{k}\lvert i-j\rvert S(i,j)$	22	25
Homogeneity	$\sum_{i=1}^{k}\sum_{j=1}^{k}\frac{S(i,j)}{1+(i-j)^2}$	17	18
Maximum Probability	$\max_{i,j}S(i,j)$	33	34
Mean	$\sum_{i=1}^{k}\sum_{j=1}^{k}S(i,j)*i$	10	21
Coefficient of variation (CV)	σ/μ, *where* σ *and* μ *are the standard deviation and mean of the samples in a given computing window, respectively.*	6	1
NRCS	N/A	8	4
Incidence angle	N/A	2	2

TABLE III

HYPERPARAMETERS OF THE RF MODEL FOR ICE–WATER CLASSIFICATION

Hyperparameters	Definition	Range	Optimal values
n_estimators	The number of trees in the forest.	100:100:1000	400
criterion	The function to measure the quality of a split. Supported criteria are "gini" for the Gini impurity and "entropy" for the information gain.	gini, entropy	entropy
max_depth	The maximum depth of the tree. If None, then nodes are expanded until all leaves are pure or until all leaves contain less than min_samples_split samples.	10:10:100, None	None
max_features	The number of features to consider when looking for the best split. If "sqrt", then max_features=sqrt(n_features). If None, then max_features=n_features.	sqrt, None	None
min_samples_split	The minimum number of samples required to split an internal node.	2, 4, 6	2
min_samples_leaf	The minimum number of samples required to be at a leaf node.	1, 2, 4	1

The basic concept behind this is that a group of "weak learners" aggregate to build a "strong learner." Compared to the conventional decision tree model, RF can generate more accurate classification results because it merges the predicted values of all decision trees. RF classifier does not require much hyperparameter tuning or feature scaling and, therefore, it is easy to use and yields a robust classification. Moreover, RF can evaluate feature importance, which gives it the potential to identify the most prominent features. More details about RF are introduced in [69]. The RF classifier has been widely used in the remote sensing community because of its ability to deal with high-dimensional feature spaces, to achieve favorable computational efficiency and highly accurate classifications [70].

In order to eliminate feature redundancy and reduce model complexity, we combine the recursive feature elimination (RFE) method with the RF model to select the most representative features for ice–water classification. RFE is an efficient method for removing dispensable features. The main idea of RFE is application of a designated classifier on subsets of features of decreasing size. In this study, we use RFE to eliminate one feature at a time. A previous study has demonstrated that better classification results are derived by eliminating one feature at a time than by excluding "chunks" of features [71]. To elaborate the procedure of feature selection, the number of features is assumed to be N.

For the first training, N features are used to train a RF model, yielding ice–water classification and feature ranking results. We discard the lowest ranking feature and use the residual $N-1$ features to train the RF model. Once again, the classifications and feature rankings are generated. The same procedure is iterated N times. When the iteration process stops, we choose a set of features corresponding to the highest classification accuracy. In this study, prior to feature selection, the input features include 33 polarimetric features (PD, cross-polarization ratio, scattering mechanism entropy, and so on) and three image features (σ^0_{HH}, σ^0_{HV}, and θ). We also apply the RFE approach to select optimal texture features from 18 estimated texture parameters, as shown in Table II. During the procedure for feature selection, we use a "10-fold shuffled split cross validation and grid search" approach to determine six hyperparameters for the RF classifier. The 10-fold shuffled split cross validation implies that the RF classifier undergoes ten rounds of training with different datasets. For each round, the input dataset is randomly divided into the training and test subsets according to a predetermined ratio. The percentages for the training and test datasets are 70% and 30%, respectively. The definition, range, and optimal values of six hyperparameters are presented in Table III.

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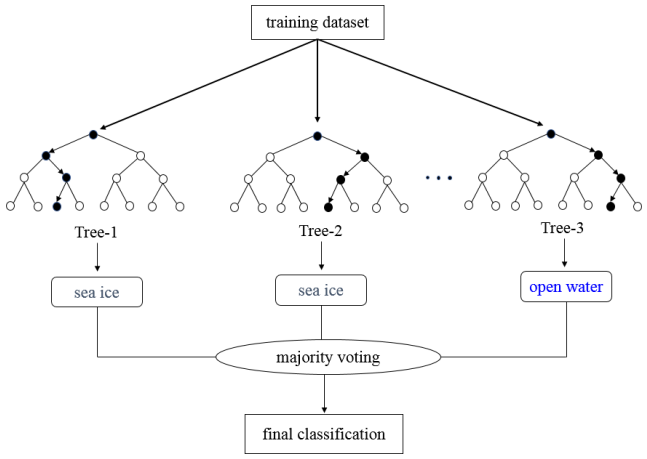


Fig. 5. General architecture of ice–water classification with RF model.

To demonstrate the importance of polarimetric features to ice–water classification, we design three classification experiments based on the RF model. The model inputs of the first experiment only include three image features, namely, HH- and HV-pol NRCS (σ^0_{HH}, σ^0_{HV}), and radar incidence angle (θ). For the second experiment, 11 optimal texture features and three image features are used as model inputs. Regarding the third experiment, 14 optimal polarimetric features and three image features are taken as the model inputs. In terms of these three experiments, the model output is the type of each pixel, whether sea ice or open water. Note that the optimal hyperparameters, the training and test datasets for all experiments are the same. The procedure for ice–water classification from the RF model consists of four steps: 1) randomly select a number of samples and features from the training dataset; 2) construct and train individual decision trees with given inputs; 3) generate the discrimination result of each decision tree; and 4) determine the final classifications based on majority voting. Fig. 5 shows the overall architecture of this methodology for the separation of sea ice and open water using the RF model. Comparisons of ice–water classification for the three experiments are presented and analyzed in Section IV.

IV. Results and Validation

A. Feature Importance

Based on the RFE method, the 17 most relevant parameters are chosen as inputs for Experiment III, including 14 optimal polarimetric features. The ranking importance of polarimetric features is summarized in Table I. This shows that the PD (VV–HH) is the foremost polarimetric parameter, with the highest ranking. PD is sensitive to wind speed [47] and is considered to be an indicator of ocean surface roughness variability. Moreover, PD does not involve the contributions from nonpolarized scattering, which can enhance the changes of surface roughness caused by surface resonant scattering [72]. These two properties enable PD to be an important polarimetric feature for separating sea ice from open water. Previous studies show that PD is less dependent on radar incidence angle than the scattering entropy and co-polarization ratio, suggesting that PD can also be used to identify newly formed sea ice in leads [73].

TABLE IV
Comparisons of Ice–water Classifications Derived From Three Experiments With Different Model Inputs

Experiment ID	Model input	Test dataset	Validation dataset	
		Overall Accuracy	Overall Accuracy	Kappa
I	σ^0_{HH}+σ^0_{HV}+θ	89.43%	89.81%	0.791
II	σ^0_{HH}+σ^0_{HV}+θ+11 texture features	96.36%	96.36%	0.925
III	σ^0_{HH}+σ^0_{HV}+θ+14 polarization features	99.93%	99.94%	0.999

Moreover, Table I shows that the cross-polarization ratio (γ_{cr}), scattering entropy (H), anisotropy ($A12$), and co-polarization ratio (γ_{co}) are also very useful for ice–water discrimination. As well, note that H and γ_{co} can be used to separate new thin ice from open water and thick sea ice [73]. In addition to these parameters, "geometric intensity" (ν) can make an important contribution to the distinguishing of sea ice and open water due to its rank importance in Table I. This parameter has previously been found to be one of the most useful variables for sea ice classification and ice–water separation [27].

In our study, we also apply the RFE approach to select optimal texture parameters. Ultimately, 11 of 18 texture features are selected as inputs for Experiment II. The ranking importance of the texture features is summarized in Table II. Among these parameters, for HV-pol, the coefficient of variation (CV) has the best ranking and is the most representative texture feature to discriminate sea ice and open water. Contrast is another texture feature that is important for HV-pol. For HH-pol, sum average is the paramount texture feature, followed by CV. These results are in line with results for feature importance presented in [53].

B. Algorithm Validation

To evaluate the performance of the proposed method for ice–water classification, we conduct both statistical validation and case-by-case comparisons. Our assessment of the classification accuracy is performed using two metrics, namely, accuracy and the Kappa coefficient. Accuracy is defined as the ratio between the number of correctly classified sea ice and open water pixels and the total number of pixels for the classification.

1) Statistical Validation: Table IV presents the comparisons of ice–water classifications derived from three experiments with different model inputs. For Experiment I, with three image features used as inputs of the RF model, the overall accuracy and the Kappa coefficient are 89.81% and 0.791, respectively. For Experiment II, these two metrics are 96.36% and 0.925, suggesting that the classification is improved by using additional texture features (CV, contrast, sum average, and so on). Finally, for Experiment III, the sea ice and open water are well separated with the highest accuracy of 99.94%

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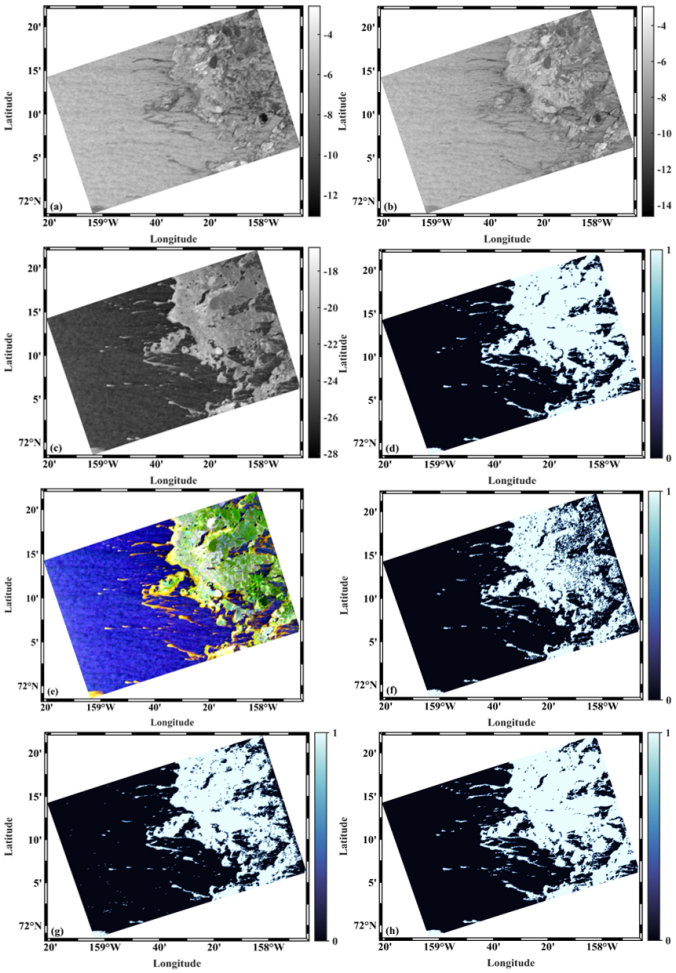


Fig. 6. (a) VV-, (b) HH-, and (c) HV-polarized RS-2 SAR images acquired over the Beaufort Sea on November 3, 2015 at 03:35 UTC. The color bar represents NRCS in VV-, HH-, and HV-polarization, in units of decibels. (d) I–water classification map ("reference") from the combined use of H/A/Alpha–Wishart classifier and visual interpretation. (e) Composite Pauli RGB image, with red (|HH + VV|), green (|HV|), and blue (|HH − VV|). (f)–(h) Ice–water classification results derived from Experiments I–III, respectively. The light cyan and dark in the color bar stand for sea ice and open water, respectively. RS-2 Data and Product MacDonald, Dettwiler, and Associates Ltd., all rights reserved.

and a Kappa coefficient of 0.999, when optimal polarimetric features (PD, cross-polarization ratio, scattering mechanism entropy, and so on) are involved in the model input. These results suggest that the proposed method can achieve relatively high accuracy for ice–water classification results. Compared to Experiments I and II, our method (Experiment III) improves ice–water discrimination accuracy by about 4%–10%. For these three experiments, the accuracy and Kappa coefficient estimated from the validation dataset are very close to those calculated from the test dataset. This implies that the RF model can derive robust ice–water classification results, no matter what model inputs are used.

2) Case Validation: Fig. 6(a)–(c), respectively, shows RS-2 VV-, HH-, and HV-pol SAR images acquired over the Beaufort Sea on November 3, 2015 at 03:35 UTC. For this case, the incidence angles are between 28.5° and 31.5°. HH-pol is better suitable for operational sea ice monitoring than VV-pol because ocean clutter is more suppressed at the former than at the latter, resulting in a higher contrast between sea ice and open water [74]. As such, HH is more appropriate for ice–water discrimination than VV. Based on a cross-polarization wind speed retrieval model [48], we estimate ocean surface wind speeds in the open water areas using the HV-pol SAR image. The average wind speed over the open water regions is 17.3 m/s. Fig. 6(a) and (b) shows that wind-roughened open water and multiyear ice-covered areas have very similar backscatter intensities. For example, the HH-pol radar returns from the sea ice areas are between −13.1 and −2.9 dB, and they range from −14.6 to −4.5 dB for the open water regions. Consequently, it is extremely difficult to discriminate open water and sea ice under high wind conditions, when only the HH- or VV-pol SAR image is used. By contrast, the HV-pol SAR image exhibits higher ice–water contrast than the HH- and VV-pol images, especially in the complicated MIZ, as shown in Fig. 6(c). The HV-pol radar returns from the sea ice areas are between −27.5 and −16.8 dB, while for open water regions, the corresponding values are between −30.5 and −23.0 dB. This high contrast characteristic facilitates the distinguishing of sea ice and open water. Moreover, compared to HH- and VV-pol, the small ice floes and leads are more clearly visible in the HV-pol SAR image.

Fig. 6(d) shows ice–water classifications from the combined use of the H/A/Alpha–Wishart classifier and visual interpretation. The H/A/Alpha–Wishart classifier is a physics-based segmentation scheme, which can be enhanced with visual interpretation of the SAR image. The high-resolution (200 m) ice–water classification map derived from this classifier can serve as a proxy of "ground truth" (or "reference") because the results exhibit extremely similar sea ice and open water patterns to those visible in the HV-pol SAR image and the composite Pauli RGB image presented in Fig. 6(e), even for small-scale features (e.g., ice floes and leads). Fig. 6(f) shows the ice–water classification map derived from Experiment I. As mentioned above, the inputs of the RF model are three image features for Experiment I. Although the overall classification accuracy is 92.75%, the classification results display notable discrepancies from the "reference" Fig. 6(d). Considerable sea ice pixels are wrongly classified as open water, especially in the far range. This is possibly caused by the significant overlap of HH-pol radar backscattering from sea ice and open water. The results of Experiment I suggest that the application of only radar backscatter intensity and incidence angle information is not sufficient to obtain reliable ice–water classifications, especially under high wind conditions. Experiment II includes the additional texture features in the model input and, therefore, derives the improved classifications shown in Fig. 6(g). The overall classification accuracy is 96.92%. However, we observe that a small quantity of open water pixels are classified as sea ice, in addition to a number of sea ice areas being identified as open water. Fig. 6(h) shows that Experiment III significantly reduces the number of misclassifications of sea ice as open water. Moreover, open water is now rarely classified sea ice. Thus, the highest classification accuracy (97.58%) is achieved for

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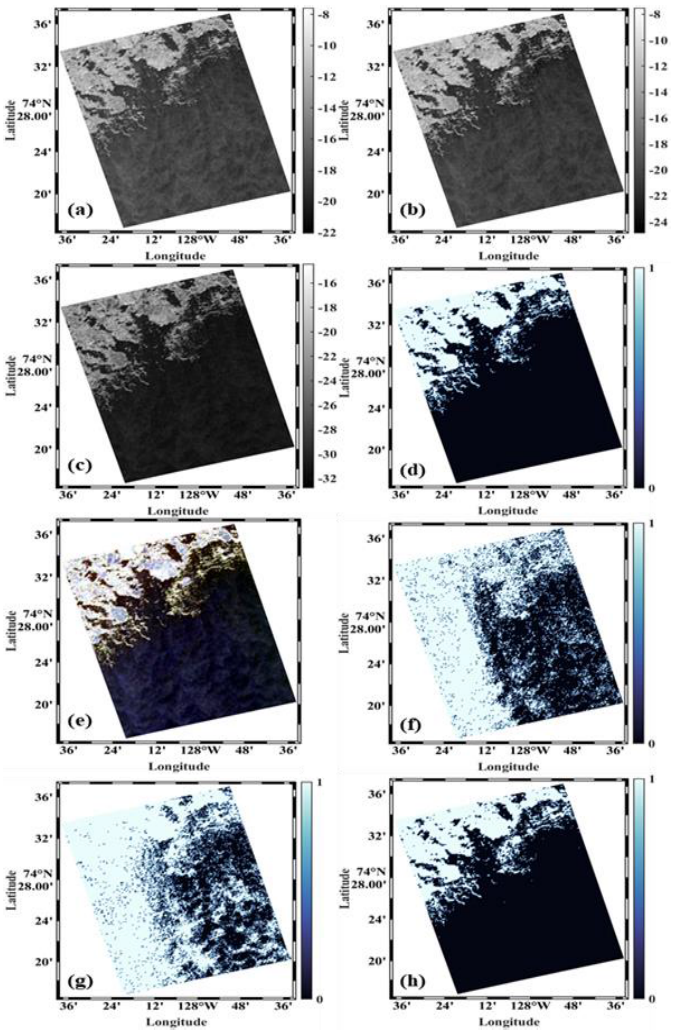


Fig. 7. (a) VV-, (b) HH-, and (c) HV-polarized RS-2 SAR images acquired over the Beaufort Sea on September 15, 2010 at 01:43 UTC. The color bar represents NRCS in VV-, HH-, and HV-polarization, in units of decibels. (d) Ice–water classification map ("reference") from the combined use of H/A/Alpha–Wishart classifier and visual interpretation. (e) Composite Pauli RGB image, with red (|HH + VV|), green (|HV|), and blue (|HH – VV|). (f)–(h) Ice–water classification results derived from Experiments I–III, respectively. The light cyan and dark in the color bar stand for sea ice and open water, respectively. RS-2 Data and Product MacDonald, Dettwiler, and Associates Ltd., all rights reserved.

Experiment III. This is due to the inclusion of the polarimetric features in the model input. For this case, the multiyear ice and open water areas are associated with different scattering mechanisms. The former is related to volume scattering, while the latter corresponds to surface scattering. Therefore, the selected polarimetric features can be used to discriminate sea ice and open water well based on the different scattering mechanisms.

Fig. 7(a)–(c), respectively, shows RS-2 VV-, HH-, and HV-pol SAR images acquired over the Beaufort Sea on September 15, 2010 at 01:43 UTC. For this case, the incidence angles range from 37.5° to 38.9°, and the average wind speed over open water area is 8.1 m/s. In the complicated MIZ, sea ice and open water have comparable HV-pol radar backscattering. Moreover, for some ice-covered areas, HV-pol radar returns are analogous to those from open water regions. This is very likely related to reduced volume scattering due to a fraction of the ice area with surface melting of the multiyear ice. Fig. 7(d) shows the referenced ice–water classification map. This reference is in good overall agreement with quad-pol SAR observations and also the composite Pauli RGB image, as shown in Fig. 7(e). The classifications from Experiments I and II are not satisfied because a large number of open water areas are misclassified as sea ice areas [see Fig. 7(f) and (g)], especially in the near range. Moreover, some sea ice areas are also misclassified as open water, due to the similar radar backscatter and textural characteristics. The classification accuracies of these two experiments are 64.14% and 60.69%. As the ice–water contrast is low, these results suggest that sea ice and open water cannot be well discriminated, even when texture features are included in the model input, at least for this case. However, for Experiment III, Fig. 7(h) shows excellent classification results derived, which resemble the reference map and SAR observations. The overall classification accuracy is 98.43%. Compared to Experiments I and II, the significant improvement obtained by Experiment III implies that polarimetric features should be considered in the SAR ice–water classification.

Fig. 8(a)–(c), respectively, shows the RS-2 VV-, HH-, and HV-pol SAR images acquired over the Beaufort Sea on June 1, 2011 at 02:28 UTC. For this case, the incidence angles are between 42.8° and 44.2°. We observe quite a few ice floes in the bottom portions of each image. Wind-driven deformation and thermal cracking may cause fragmentation of the ice pack into floes. These ice floes manifest various morphology features because of the combined action of wind, waves, and currents. On the top-left corner of each panel, especially in the HH-pol SAR image as shown in Fig. 8(b), a large area of pack ice can be found. Due to the effect of surface melting, HV-pol backscattering from the multiyear sea ice is not much higher than that from open water areas. Fig. 8(d) and (e) present the reference map of ice–water classification and the composite Pauli RGB image, respectively. For this case, Experiment I misclassifies a lot of sea ice areas as open water, as shown in Fig. 8(f), with an overall accuracy of 75.42%. Fig. 8(g) shows the classification results from Experiment II. The accuracy (75.27%) of Experiment II is slightly higher than that of Experiment I. Compared to Experiment I, the number of sea ice misclassifications is significantly reduced, but considerable open water areas are classified as sea ice. Finally, as in previous examples, Experiment III derives the best ice–water classification results with the highest accuracy of 91.48%. Fig. 8(h) shows that the classification map of Experiment III is more consistent with the reference map and the SAR observations, compared to Experiments I and II. This case indicates that the proposed method has the potential to identify sea ice and open water well.

Fig. 9(a)–(c), respectively, shows RS-2 VV-, HH-, and HV-pol SAR images acquired over the Beaufort Sea on October 8, 2010 at 02:12 UTC. For this case, the incidence angles range from 42.9° to 44.1°. Due to the high-resolution (∼5 m) of quad-polarization SAR, several long, narrow leads are clearly visible in ice pack regions, with very low radar backscattering. HH- and VV-pol SAR images exhibit better

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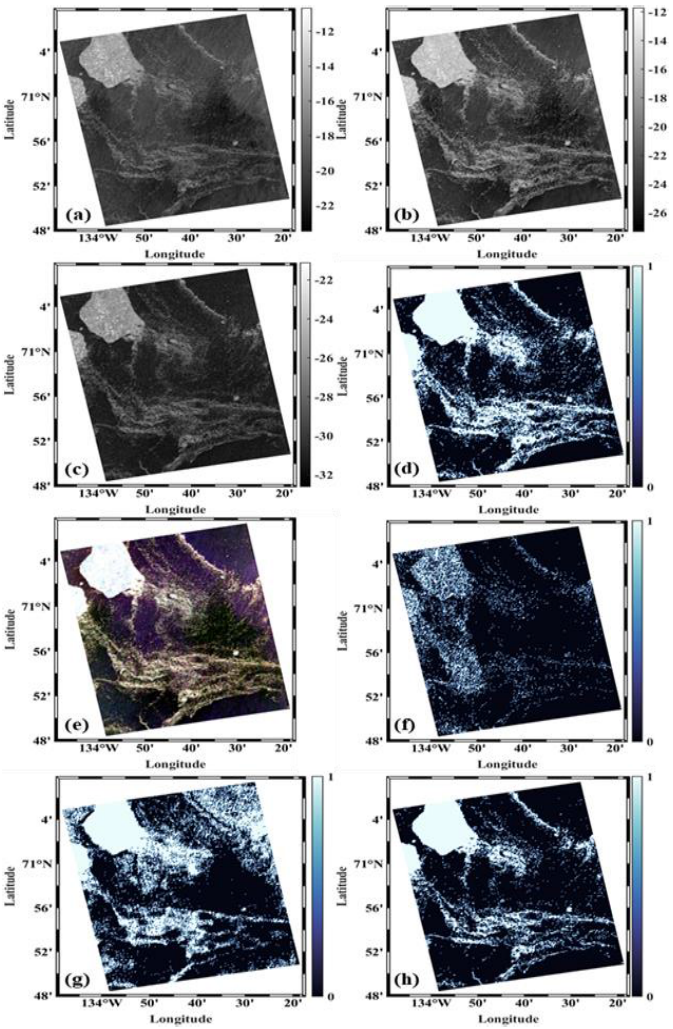


Fig. 8. (a) VV-, (b) HH-, and (c) HV-polarized RS-2 SAR images acquired over the Beaufort Sea on June 1, 2011 at 02:28 UTC. The color bar represents NRCS in VV-, HH-, and HV-polarization, in units of decibels. (d) Ice–water classification map ("reference") from the combined use of H/A/Alpha–Wishart classifier and visual interpretation. (e) Composite Pauli RGB image, with red (|HH + VV|), green (|HV|), and blue (|HH − VV|). (f)–(h) Ice–water classification results derived from Experiments I–III, respectively. The light cyan and dark in the color bar stand for sea ice and open water, respectively. RS-2 Data and Product MacDonald, Dettwiler, and Associates Ltd., all rights reserved.

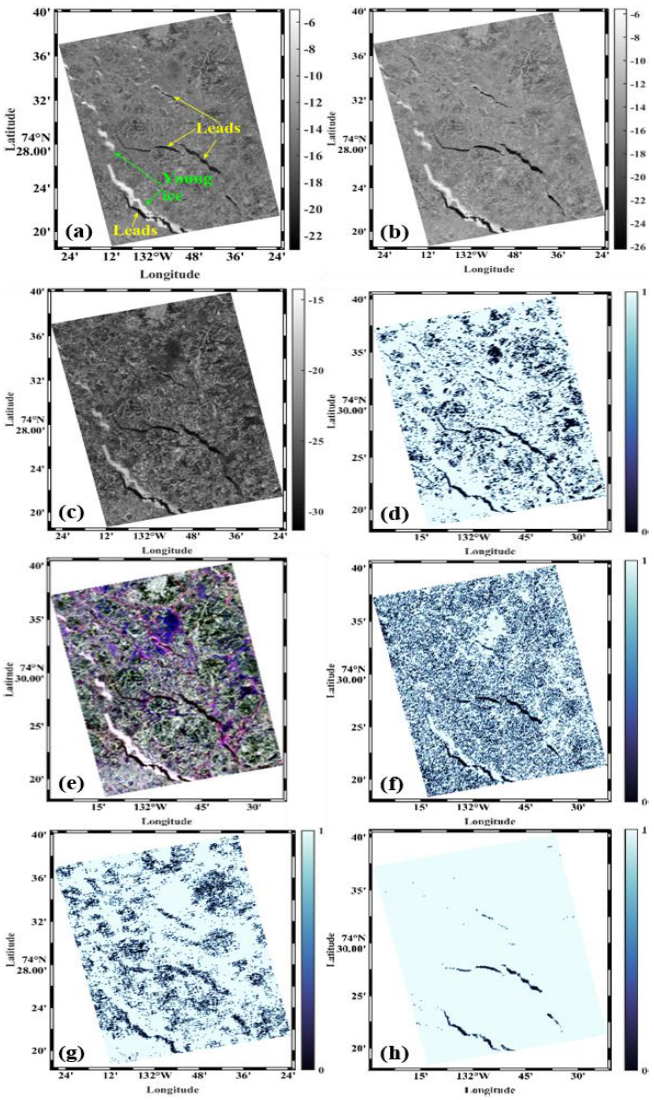


Fig. 9. (a) VV-, (b) HH-, and (c) HV-polarized RS-2 SAR images acquired over the Beaufort Sea on October 8, 2010 at 02:12 UTC. The color bar represents NRCS in VV-, HH-, and HV-polarization, in units of decibels. (d) Ice–water classification map ("reference") from the combined use of H/A/Alpha–Wishart classifier and visual interpretation. (e) Composite Pauli RGB image, with red (|HH + VV|), green (|HV|), and blue (|HH − VV|). (f)–(h) Ice–water classification results derived from Experiments I–III, respectively. The light cyan and dark in the color bar stand for sea ice and open water, respectively. RS-2 Data and Product MacDonald, Dettwiler, and Associates Ltd., all rights reserved.

contrast between sea ice and leads than HV-pol. As shown in Fig. 9(c), HV-pol radar backscatters from several ice-covered areas are similar to backscatters from the leads, which indicates that surface scattering is possibly dominant in these areas. Fig. 9(d) shows that sea ice in these regions is misclassified as open water areas, even for the H/A/Alpha–Wishart classifier; this is demonstrated by visual inspection of HH- and VV-pol SAR images as well as the composite Pauli RGB image presented in Fig. 9(e).

Results from Experiment I are shown in Fig. 9(f), which indicates that a large number of sea ice areas are misclassified as open water. Fig. 9(g) shows that Experiment II experiences the similar problem as Experiment I. Moreover, Experiment II classifies a portion of ice leads as sea ice. Experiment III identifies sea ice and leads very well and generates more reasonable classification results than Experiment I or II, as shown in Fig. 9(h). We also observe young ices in the VV-, HH-, and HV-pol SAR images, which are identified as bright linear features in Fig. 9(a)–(c). Note that Experiment III does not classify these young ices as open water. However, several pieces of young ice are classified as open water by Experiments I and II. For this case, we do not estimate the overall classification accuracy of the three experiments because the reference shown in Fig. 9(d) cannot be considered as a perfect "ground truth."

Conventional methods rely on visual interpretation of SAR images and manually derived ice charts by ice forecast experts, to create annotated samples. Thus, the labeled samples are used to train machine learning models for classifying sea ice

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and open water. Here, different from traditional approaches, we apply the H/A/Alpha–Wishart classifier to automatically segment RS-2 quad-pol SAR images and to achieve unsupervised generation of the training samples. The H/A/Alpha classifier is first used to implement initial SAR image classification because entropy (H), alpha angle (α), and anisotropy (A) are related to the implicit scattering process (H, A) and physical scattering mechanism (α). We further use the Wishart classifier to ultimately determine the specific class of each pixel based on estimated minimum distance. It should be noted that the H/A/Alpha–Wishart classifier may yield inaccurate segmentation results when scattering characteristics of sea ice and open water are very similar. Thus, smooth first-year sea ice areas may be misclassified as open water by the H/A/Alpha–Wishart classifier, as shown in Fig. 9(d). To obtain high-quality training samples, we exclude erroneous classifications derived from this classifier by inspecting HH- and HV-pol SAR images and the associated composite Pauli RGB image. Despite this, the H/A/Alpha–Wishart classifier may still generate some incorrect classifications and thus affect the accuracy of ice–water classification results.

H, A, and α are important polarimetric features and play important roles in initial classification. Compared to H, A, and α, PD and cross-polarization ratio (γ_{cr}) have much higher rank of importance (Table I), indicating that PD and γ_{cr} are more important for discriminating sea ice from open water. We have presented the detailed explanations in Section IV-A. Moreover, A is not selected as an optimal polarimetric feature and is not taken as model input due to its very low-rank importance. Note that the inputs of the RF model not only include the most favorable polarimetric features but also image features. HV-pol radar backscatter is an important model input because of strong contrast between sea ice and open water, especially in the complicated MIZ. This image feature is ranked second in terms of feature importance. Thus, the performance of fully polarimetric SAR ice–water classification is not strongly dependent on H, A, and α. It is clear that to further improve the model performance, a more comprehensive and balanced dataset needs to be created.

V. Summary and Conclusion

In this study, we propose a physics-based and data-driven method to automatically classify sea ice and open water for spaceborne C-band fully polarimetric SAR. A physics-based image classification scheme, called the H/A/Alpha–Wishart classifier, is used for unsupervised segmentation of sea ice and open water. By comparing the segmentation results with quad-pol SAR imagery, the ice–water classification map (ground truth) is generated. We estimate a set of polarimetric features using fully polarimetric SAR scattering matrix elements and eigendecomposition parameters of the coherence matrix. The optimal polarimetric features are selected using the RFE method and the RF model. We find that PD (VV–HH) is the foremost polarimetric parameter for distinguishing sea ice from open water. The most representative 14 polarimetric features (PD, cross-polarization ratio, scattering mechanism entropy, and so on), along with three image features (HH- and HV-pol NRCS, incidence angle), are collocated with the ice–water classification reference map, generating the training, test, and validation samples. A total of 1 311 609 labeled samples are used to train and validate the RF model. We conduct statistical validations to evaluate the proposed method and the results show that the overall classification accuracy is 99.94%. Based on the same datasets, we also use the RF model to classify sea ice and open water, with different inputs. For example, model inputs are three image features or the combination of three image features and 11 texture features (CV, contrast, sum average, and so on). The classification accuracies are 89.81% and 96.36%. These results suggest that ice–water discrimination accuracy can be improved by about 4%–10% when polarimetric features are considered.

In addition to a statistical assessment, we also carry out case-by-case validations using fully polarimetric SAR imagery acquired in different seasons under different wind speeds. The classification accuracies of four cases are greater than 90%, even in the melting season. Note that the SAR images corresponding to these cases are not included in the training and test datasets. Under high wind conditions (~17 m/s), the proposed method is still able to separate sea ice and open water very well, with a high accuracy of 97.58%. The case classification map also shows that small-scale features (e.g., leads and ice floes) can be identified using our ice–water classification approach. Moreover, sea ice and open water are properly discriminated in the complicated MIZ. From the perspective of operational sea ice monitoring from space, wide swath SAR observations are required. Although we have demonstrated that fully polarimetric SAR data provide abundant polarimetric information and facilitate our ability to distinguish sea ice and open water, the area coverage is much smaller compared to single- or dual-polarization SAR. Fortunately, HRWS SAR systems [35], [36], [37], [38], [39] have been designed and may provide fully polarimetric observations over large areas. Thus, the proposed ice–water classification method has the potential to be used in the future operational sea ice monitoring of the Arctic.

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